

Omin Kwon

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RESEARCH INTEREST

Computer Architectures & Machine Learning Systems, with an emphasis on co-designing ML algorithms and computer systems for efficient ML.

EDUCATION

Seoul National University

Sept 2025 – present

M.S. in Computer Science and Engineering

- Advisor: Prof. Jae W. Lee, ARC Lab
- GPA: 4.3/4.3

Seoul National University

Mar 2019 – Aug 2025

B.S. in Electrical and Computer Engineering (summa cum laude)

- GPA: 4.26/4.3 (upper-division), 3.95/4.3 (overall)
- Leave of absence for mandatory military service (May 2021-Feb 2023)

Hansung Science High School

Mar 2017 – Mar 2019

H.S. in Science

PUBLICATIONS

SketchSSM: Write to the Full State, Read from a Compact Sketch

Omin Kwon, JoongWon Shin, Minseo Kim, Kurt Keutzer, Sehoon Kim, Jae W. Lee
Under review

HERALD: High Throughput Block Diffusion LLM Serving via CPU-GPU Cooperative KV Cache Retrieval

Omin Kwon, Doyeon Kim, Jongseok Park, Seung Yul Lee, Ion Stoica, Jae W. Lee
Under review

ELIXIR: Disaggregating FFN, Attention, and KV Cache via CXL-PNM for Elastic LLM Serving

Omin Kwon*, KyungSoo Kim*, Haneul Park, JoongWon Shin, Nachuan Wang, UGyeong Song, Yeonhong Park, Nam Sung Kim, Jae W. Lee
Under review

MAGE: All-[MASK] Block Already Knows Where to Look in Diffusion LLM

Omin Kwon, Yeonjae Kim, Doyeon Kim, Minseo Kim, Yeonhong Park, Jae W. Lee
NeurIPS 2026; ICML 2026 AdaptFM Workshop (Best Paper Award)

STAR: Improving Lifetime and Performance of High-Capacity Modern SSDs Using State-Aware Randomizer

Omin Kwon*, Kyungjun Oh*, Jaeyong Lee, Myungsuk Kim, Jihong Kim
ICCAD 2025

NestedFP: High-Performance, Memory-Efficient Dual-Precision Floating Point Support for LLMs

Haeun Lee, **Omin Kwon**, Yeonhong Park, Jae W. Lee
NeurIPS 2025

AiDE: Attention-FFN Disaggregated Execution for Cost-Effective LLM Decoding on CXL-PNM

KyungSoo Kim, **Omin Kwon**, Yeonhong Park, Jae W. Lee
IEEE Computer Architecture Letters (CAL)

EXPERIENCES

Teaching Assistant, Advanced Computer Architecture (4190.571) Seoul National University	Sept 2026 – present Seoul, South Korea
Visiting Researcher University of Illinois Urbana-Champaign (UIUC) • FAST Lab, hosted by Prof. Nam Sung Kim	Mar 2026 – May 2026 Urbana, IL, USA
Microelectronics (430.207B) Tutor Seoul National University	Mar 2025 – June 2025 Seoul, South Korea
Digital System Design (430.315) Tutor Seoul National University	Sept 2024 – Dec 2024 Seoul, South Korea
Computer Organization and Design (430.322) Tutor Seoul National University	Mar 2024 – June 2024 Seoul, South Korea
Military Service Republic of Korea Air Force	May 2021 – Feb 2023 South Korea

PRESENTATIONS

MAGE: All-[MASK] Block Already Knows Where to Look in Diffusion LLM Oral presentation at the ICML 2026 AdaptFM Workshop. arXiv.	2026
STAR: Improving Lifetime and Performance of High-Capacity Modern SSDs Using State-Aware Randomizer Oral presentation at ICCAD 2025, Munich. Paper.	2025

HONORS & AWARDS

Best Paper Award @ ICML26 AdaptFM Workshop ICML 2026 AdaptFM Workshop 1 Best Paper among 123 accepted papers from 218 submissions.	July 2026
Grand Prize (Minister of Science and ICT Award), 2024 AI Semiconductor Idea and Design Contest Ministry of Science and ICT Designed a resolution-aware diffusion accelerator leveraging computational staleness. News: Edaily.	Jan 2025
Grand Prize (Deputy Prime Minister and Minister of Education Award), 2024 STOB League Semiconductor Contest Ministry of Education Designed a state-aware data randomizer for high-performance SSDs, improving performance and device lifetime. News: Dong-A Ilbo.	Dec 2024
Grand Prize, 2024 FriendliAI LLM Hackathon FriendliAI Developed a knowledge-graph RAG application using the FriendliAI API.	July 2024
SNU Specialized Semiconductor College Scholarship Seoul National University	Aug 2024 – Aug 2025
Merit-Based Scholarship Seoul National University	2023, 2024